Science practice booklet: Grade 6 expectations

Grade 6 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 6 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: MS-LS1-1, MS-LS1-2, MS-LS1-3, MS-LS1-6, MS-LS2-1, MS-LS2-3

A worked example

A mangrove leaf is made of cells. Name the part of a leaf cell that captures light for photosynthesis, and say what the leaf makes and gives off.

1. Plant cells have green parts called chloroplasts. They contain a green substance that captures light energy.
2. In photosynthesis the plant uses light energy, water and carbon dioxide.
3. These are changed into sugar (food for the plant) and oxygen.
4. The sugar is used or stored by the plant, and the oxygen is released into the air.

Chloroplasts capture the light; the leaf makes sugar and gives off oxygen.

Questions

1. What is the basic unit of structure and function in all living things?

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2. What is the job of the cell membrane?

   A. It makes food for the cell

   B. It controls what goes into and out of the cell

   C. It holds the instructions for the cell

   D. It releases energy from food

3. Which part of a cell holds the instructions that control the cell's activities?

   A. The nucleus

   B. The cell membrane

   C. The vacuole

   D. The mitochondria

4. Which part is found in plant cells but not in animal cells?

   A. The nucleus

   B. The mitochondria

   C. The cell membrane

   D. The chloroplasts

5. Mitochondria are called the 'powerhouses' of the cell. What do they do?

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6. Which two things does a plant take in for photosynthesis?

   A. Oxygen and sugar

   B. Oxygen and water

   C. Carbon dioxide and water

   D. Sugar and carbon dioxide

7. Put these in order from smallest to largest: organ, cell, organ system, tissue.

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8. Match each body system to its job: circulatory, respiratory, digestive. (Jobs: breaks down food; brings oxygen in; carries blood around the body.)

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9. Hamdan runs in the school yard and his breathing and heartbeat both get faster. Explain how two body systems work together here.

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10. In a mangrove food web, plants make food, crabs eat leaves and herons eat crabs. Which organisms have the most energy available, and why does the amount get smaller along the chain?

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11. Oryx and gazelles in a desert reserve share a small number of water holes and plants. What can happen if a very dry year reduces the plants?

    A. There is less food, so the animals compete more and their numbers may get smaller

    B. There is more food, so their numbers grow

    C. Nothing changes, because animals do not need plants

    D. The oryx move into the sea to find food

12. A seedling is kept in a dark cupboard for two weeks and turns pale and droopy. Use photosynthesis to explain why.

Section 2: Earth and space

Standards practised: MS-ESS2-1, MS-ESS2-2, MS-ESS2-3, MS-ESS2-4, MS-ESS2-6

A worked example

Seawater in the Gulf warms in the sun, and later rain falls on the Hajar mountains. Describe the steps of the water cycle that connect these two events and say what powers the cycle.

1. The Sun warms the sea and some of the water evaporates, turning into water vapour (a gas).
2. The warm air rises and cools. The vapour condenses into tiny drops that form clouds.
3. When the drops get big enough they fall as precipitation, such as rain.
4. The Sun's energy and gravity keep the water moving around the cycle.

Evaporation, condensation, then precipitation; the Sun's energy and gravity power the cycle.

Questions

1. Name the three main steps of the water cycle.

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2. Water drops form on the outside of a cold glass of juice on a humid day. Which step of the water cycle is this like?

   A. Evaporation

   B. Precipitation

   C. Condensation

   D. Melting

3. Which two things move water around the water cycle?

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4. Which statement is about climate and not about weather?

   A. Summers in the UAE are hot and dry year after year

   B. It rained in Fujairah this morning

   C. Tomorrow will be dusty and windy

   D. Al Ain reached 41 degrees today

5. Maryam writes down the weather in her notebook every day for one month. Is her notebook a record of weather or of climate? What else would she need to describe the climate?

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6. List Earth's four main layers from the outside to the centre.

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7. Which layer of the Earth do we live on, and what is it like?

   A. The mantle, the thickest layer

   B. The crust, the thinnest layer

   C. The outer core, a liquid layer

   D. The inner core, a solid layer

8. Earth's crust is broken into large pieces that move a few centimetres each year. What are these pieces called, and name one event they can cause where they meet.

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9. Scientists find sea shell fossils high in the Hajar mountains. What does this suggest?

   A. The mountains have always looked the same

   B. Sea animals climbed up the mountains

   C. The wind made the shells inside the rock

   D. The place was once under the sea and the land was later lifted up

10. On a walk with his cousins, Khalid finds three rocks: one made of crystals that formed when melted rock cooled, one with thin layers of pressed sand, and one that was changed by great heat and pressure. Name the type of each rock.

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11. Wind blows sand and builds a dune. Over millions of years the layers of sand are pressed and cemented into rock. Which type of rock forms?

    A. Igneous rock, made when melted rock cools

    B. Metamorphic rock, made when heat and pressure change older rock

    C. Sedimentary rock, made when layers of sediment are pressed together

    D. Fossil rock, which is a fourth main type

12. Explain why the rock cycle never ends. Give an example of one rock changing into another type.

Section 3: Matter, forces and energy

Standards practised: MS-PS1-4, MS-PS3-3, MS-PS3-4, MS-PS2-2

A worked example

Noura puts an ice cube in a cup in a warm kitchen. Describe what happens to the particles of water as the ice melts and then the water warms up.

1. In ice (a solid) the particles are packed closely and only vibrate in place.
2. Heat flows from the warm room into the ice, and the particles gain energy and move more.
3. They loosen and slide past each other: the ice melts into liquid water.
4. As the liquid keeps getting heat, the particles move faster and the temperature rises.

The particles gain energy, move more and slide past each other (melting), then move faster as the water warms.

Questions

1. Describe how the particles are arranged and move in a solid, a liquid and a gas.

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2. What happens to the particles of a substance when it is heated?

   A. They get bigger

   B. They gain energy and move faster

   C. They stop moving

   D. They move slower and closer together

3. At what temperature does pure water boil and at what temperature does it freeze, in degrees Celsius?

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4. A puddle dries up in the sun. Which change of state has happened?

   A. Condensation

   B. Melting

   C. Freezing

   D. Evaporation

5. A football left in a hot car in the afternoon sun feels harder and tighter. Use particles to explain why.

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6. Khalid holds a hot cup of karak. Which statement about the heat is correct?

   A. Heat flows from the hot cup to his cooler hands

   B. Heat flows from his hands to the cup

   C. Cold flows from his hands into the cup

   D. No heat moves between them

7. The sand on a UAE beach is too hot to walk on at noon. How does the Sun's energy reach the sand, and which way of heat transfer is it: conduction, convection or radiation?

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8. Which pairing of an example with a way heat moves is correct?

   A. The Sun warming the sand: conduction

   B. Hot soup warming a metal spoon: convection

   C. Warm air rising above a heater: convection

   D. A hot cup warming hands: radiation

9. Hind wants her cold drink to stay cold on a hot day. Choose a material for the container and explain your choice.

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10. A bathtub of water and a cup of water are both at 30 °C. Which has more thermal energy?

    A. The cup, because it is smaller

    B. The bathtub, because it has many more particles

    C. They are the same, because the temperature is the same

    D. The cup, because its particles move faster

11. Solar panels in a UAE solar park change the Sun's light energy into another form of energy. Which form?

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12. Two teams pull on a rope with equal and opposite forces and the rope does not move. Are the forces balanced or unbalanced? What would happen if one team pulled harder?

Answer guide for parents

Section 1: Living things

1. The cell, because every living thing is built from cells and each cell does the jobs of life.
2. B. It controls what goes into and out of the cell. The membrane is the cell's outer skin. It lets in water and food and lets out waste. If your child chose C, they mixed it up with the nucleus.
3. A. The nucleus. is the control centre and holds the instructions. If your child chose D, they mixed it up with the mitochondria, which release energy.
4. D. The chloroplasts. Chloroplasts capture light and are only in plant cells. If your child chose A, they forgot that animal cells have a nucleus too.
5. Mitochondria release energy from food, like a small power station. The cell then uses that energy.
6. C. Carbon dioxide and water. Plants take in carbon dioxide and water, and use light to make sugar and oxygen. If your child chose A, they listed what the plant makes, not what it takes in.
7. Start with the smallest, a cell. Cells make a tissue, tissues make an organ, and organs make an organ system.
8. Match by the key word. Blood goes with circulatory, oxygen with respiratory and food with digestive.
9. Running makes his muscles need more oxygen. Faster breathing brings in more oxygen, and a faster heartbeat carries it to the muscles.
10. The mangrove plants are producers, so they have the most energy. Each animal uses most of its energy to live and loses some as heat, so less passes on.
11. A. There is less food, so the animals compete more and their numbers may get smaller. Less food means more competition, so animal numbers may fall. If your child chose C, they forgot that animals depend on plants for food.
12. Without light there is no photosynthesis, so the plant cannot make sugar for energy and growth. Its leaves lose their green colour too.

Section 2: Earth and space

1. Water evaporates into vapour, the vapour condenses into clouds, and then water falls as precipitation. That is the three steps in order.
2. C. Condensation. Vapour in the air cools on the cold glass and turns to liquid, which is condensation. If your child chose A, they think the water came from the juice, but it came from the air.
3. The Sun's heat lifts water into the air as vapour, and gravity pulls the rain back down. So it is the Sun's energy and gravity.
4. A. Summers in the UAE are hot and dry year after year. Climate is the usual pattern over many years, and hot dry summers year after year is a pattern. If your child chose D, they picked one day's temperature, which is weather.
5. She records one month of daily weather, so her notebook shows weather. Climate needs the pattern across many years.

6. Start at the surface we live on and go down to the centre: crust, mantle, outer core, inner core.
7. B. The crust, the thinnest layer. We live on the crust, which is the thinnest layer, like the skin of an apple. If your child chose A, they picked the thickest layer, which lies far below us.
8. The crust is broken into pieces called tectonic plates. Where they meet they can cause earthquakes, volcanoes or mountains. (Any one.)
9. D. The place was once under the sea and the land was later lifted up. Shells form in the sea, so the rock was once under water and was later lifted up. If your child chose B, they think animals climbed there, but fossils form where the animals died.
10. Cooled melted rock is igneous, pressed layers of sand are sedimentary, and rock changed by heat and pressure is metamorphic. Match each clue to a type.
11. C. Sedimentary rock, made when layers of sediment are pressed together. Layers of sand pressed and cemented together make sedimentary rock. If your child chose A, they mixed it up with igneous rock, which needs melted rock.
12. Rocks keep being worn away, melted, pressed or heated, so they keep changing type. For example, igneous rock breaks into sand, and pressed sand becomes sedimentary rock.

Section 3: Matter, forces and energy

1. Solids are packed and vibrate, liquids are close but slide, and gases are far apart and move freely. More space between particles means more freedom to move.
2. B. They gain energy and move faster. Heat gives particles energy, so they move faster. If your child chose A, they think particles grow when heated, but only their speed changes.
3. Pure water freezes at 0 °C and boils at 100 °C, so the answer is 100 °C and 0 °C.
4. D. Evaporation. The liquid water turns into vapour in the air, which is evaporation. If your child chose A, they mixed it up with condensation, where vapour turns into liquid.
5. Heat gives the air particles more energy, so they move faster. They hit the inside of the ball harder and more often, which pushes it tighter.
6. A. Heat flows from the hot cup to his cooler hands. Heat flows from hotter to colder, so it moves from the cup into his hands. If your child chose C, they think cold moves, but cold is just heat leaving.
7. The Sun's energy crosses space as light and heat rays without touching anything. That way of heat moving is radiation.
8. C. Warm air rising above a heater: convection. Warm air rises above a heater and carries heat with it, which is convection. If your child chose A, they mixed up the Sun's rays, which are radiation, with conduction by touch.
9. Pick an insulator, such as a foam box or a vacuum flask. It slows the heat flowing in from the hot air, so the drink stays cold.
10. B. The bathtub, because it has many more particles. The same temperature means the same average speed, but the bathtub has far more particles, so more thermal energy. If your child chose C, they mixed up temperature with total thermal energy.
11. Solar panels change light energy into electrical energy, which is electricity. So the answer is electrical energy.
12. Equal and opposite forces cancel out, so they are balanced and the rope stays still. If one team pulls harder, the forces are unbalanced and the rope moves towards that team.